

Incalmo: cyber red-teaming via Constrained Creativity



Task

"Gain access and exfiltrate sensitive data."



DSL / Playbook

scan

exploit

lateral move

...



LLM

plan a sequence of high-level attack actions



DSL program

```
1 Scan(hosts)
2 LateralMove(target)
3 ExfiltrateData(path)
```



DSL runtime engine

nmap

Metasploit

state tracking



Answer



foothold



pivot



find data



exfiltrate

iterate